complete java class code example of each combination of intermediate and terminal operations

filter() boolean checks + collect() map → reduce → store

filter() null checks + collect() map → reduce → store

filter() comparisons + collect() map → reduce → store

filter() method calls returning boolean + collect() map → reduce → store

distinct() rely on equals() / hashcode() + collect() map → reduce → store

distinct() immutable objects + collect() map → reduce → store

map() transform objects + collect() map → reduce → store

map() create new objects + collect() map → reduce → store

map() call pure functions + collect() map → reduce → store

flatmap() return stream + collect() map → reduce → store

flatmap() split data + collect() map → reduce → store

flatmap() expand collections + collect() map → reduce → store

sorted() natural ordering + collect() map → reduce → store

sorted() custom comparator + collect() map → reduce → store

sorted() multi-field compare + collect() map → reduce → store

peek() logging + collect() map → reduce → store

peek() debugging + collect() map → reduce → store

peek() metrics + collect() map → reduce → store

Below are **all 18 programs converted to Java 1.5 compatible code**.

Since Java 1.5:

- ❌ No Stream API
- ❌ No Optional
- ❌ No lambda expressions
- ❌ No method references
- ❌ No List.of(), Map.of()
- ❌ No Objects.hash()
- ❌ No Comparator.comparing()

We replace everything with:

- ✔ for-each loops
- ✔ Iterator
- ✔ Collections / Arrays
- ✔ HashSet for distinct
- ✔ Manual sorting with Comparator
- ✔ Null checks
- ✔ Primitive counters instead of AtomicInteger

Below are **complete, standalone Java class examples** where **each pipeline explicitly follows**:

Map → Reduce → Store

- **Map** = transform data
- **Reduce** = aggregate (sum / count / max / min / join / etc.)
- **Store** = collect() into a Map or List

All examples:

- Java **8+**
- One class = one concept
- Interview-safe and production-realistic
- No shortcuts, no magic

---

1 filter() boolean checks + collect() (Map → Reduce → Store)

```
import java.util.*;

public class FilterBooleanMapReduceStore {
   public static void main(String[] args) {

      List<Integer> numbers = Arrays.asList(10, 15, 20, 25);
      Map<String, Integer> result = new HashMap<String, Integer>();

      for (Integer n : numbers) {

         if (n % 2 == 0) {                                    // FILTER

            int mapped = n * 2;                               // MAP

            String key = (mapped > 30) ? "HIGH" : "LOW";

            Integer sum = result.get(key);
            if (sum == null) sum = 0;

            result.put(key, sum + mapped);                    // REDUCE + STORE
         }
      }

      System.out.println(result);
   }
}
```

1 filter() boolean checks + collect() (Map → Reduce → Store)

```
import java.util.*;
import java.util.stream.*;

public class FilterBooleanMapReduceStore {
   public static void main(String[] args) {
      List<Integer> numbers = List.of(10, 15, 20, 25);

      Map<String, Integer> result =
            numbers.stream()
                  .filter(n -> n % 2 == 0)                          // filter (boolean)
                  .map(n -> n * 2)                                  // MAP
                  .collect(Collectors.groupingBy(
                        n -> n > 30 ? "HIGH" : "LOW",
                        Collectors.summingInt(Integer::intValue) // REDUCE
                  ));                                               // STORE

      System.out.println(result);
   }
}
```

---

2 filter() null checks + collect() (Map → Reduce → Store)

```
import java.util.*;

public class FilterNullMapReduceStore {
   public static void main(String[] args) {

      List<String> names = Arrays.asList("Anurag", null, "Amit", null);
      Map<Integer, Long> result = new HashMap<Integer, Long>();

      for (String name : names) {

         if (name != null) {                                        // FILTER

            String mapped = name.toUpperCase();                     // MAP
            Integer key = mapped.length();

            Long count = result.get(key);
            if (count == null) count = 0L;

            result.put(key, count + 1);                             // REDUCE + STORE
         }
      }

      System.out.println(result);
```

```
  }
}
```

2 filter() null checks + collect() (Map → Reduce → Store)

```
import java.util.*;
import java.util.stream.*;

public class FilterNullMapReduceStore {
  public static void main(String[] args) {
    List<String> names = Arrays.asList("Anurag", null, "Amit", null);

    Map<Integer, Long> result =
        names.stream()
            .filter(Objects::nonNull)                    // null check
            .map(String::toUpperCase)                    // MAP
            .collect(Collectors.groupingBy(
                String::length,
                Collectors.counting()                    // REDUCE
            ));                                          // STORE

    System.out.println(result);
  }
}
```

---

3 filter() comparisons + collect() (Map → Reduce → Store)

```
import java.util.*;

public class FilterComparisonMapReduceStore {
  public static void main(String[] args) {

    List<Integer> salaries = Arrays.asList(20000, 40000, 60000);
    Map<String, Integer> result = new HashMap<String, Integer>();

    for (Integer s : salaries) {

      if (s >= 30000) {                                  // FILTER

        int mapped = s + 5000;                           // MAP
        String key = (mapped >= 50000) ? "SENIOR" : "MID";

        Integer sum = result.get(key);
        if (sum == null) sum = 0;

        result.put(key, sum + mapped);                   // REDUCE + STORE
      }
    }

    System.out.println(result);
```

```
    }
}
```

3 filter() comparisons + collect() (Map → Reduce → Store)

```
import java.util.*;
import java.util.stream.*;

public class FilterComparisonMapReduceStore {
    public static void main(String[] args) {
        List<Integer> salaries = List.of(20000, 40000, 60000);

        Map<String, Integer> result =
            salaries.stream()
                .filter(s -> s >= 30000)                    // comparison
                .map(s -> s + 5000)                         // MAP
                .collect(Collectors.groupingBy(
                    s -> s >= 50000 ? "SENIOR" : "MID",
                    Collectors.summingInt(Integer::intValue) // REDUCE
                ));

        System.out.println(result);
    }
}
```

---

4 filter() method call boolean + collect() (Map → Reduce → Store)

```
import java.util.*;

public class FilterMethodMapReduceStore {

   static class User {
      String name;
      boolean active;

      User(String name, boolean active) {
         this.name = name;
         this.active = active;
      }

      boolean isActive() {
         return active;
      }
   }

   public static void main(String[] args) {

      List<User> users = Arrays.asList(
            new User("Anurag", true),
            new User("Amit", true),
            new User("Ravi", false)
      );

      Map<Character, Long> result = new HashMap<Character, Long>();

      for (User u : users) {

         if (u.isActive()) {                                    // FILTER

            String mapped = u.name;                             // MAP
            Character key = mapped.charAt(0);

            Long count = result.get(key);
            if (count == null) count = 0L;

            result.put(key, count + 1);                         // REDUCE + STORE
         }
      }

      System.out.println(result);
   }
}
```

4 filter() method call boolean + collect() (Map → Reduce → Store)

```
import java.util.*;
import java.util.stream.*;

public class FilterMethodMapReduceStore {

   static class User {
      String name;
      boolean active;

      User(String name, boolean active) {
         this.name = name;
         this.active = active;
      }

      boolean isActive() {
         return active;
      }
   }

   public static void main(String[] args) {
      List<User> users = List.of(
            new User("Anurag", true),
            new User("Amit", true),
            new User("Ravi", false)
      );

      Map<Character, Long> result =
            users.stream()
                  .filter(User::isActive)                    // method call
                  .map(u -> u.name)                          // MAP
                  .collect(Collectors.groupingBy(
                        name -> name.charAt(0),
                        Collectors.counting()                // REDUCE
                  ));

      System.out.println(result);
   }
}
```

---

5 distinct() equals/hashCode + collect() (Map → Reduce → Store)

```
import java.util.*;

public class DistinctEqualsMapReduceStore {

   static class Person {
      String name;

      Person(String name) { this.name = name; }

      public boolean equals(Object o) {
         if (o instanceof Person) {
            Person p = (Person) o;
            return name.equals(p.name);
         }
         return false;
      }

      public int hashCode() {
         return name.hashCode();
      }
   }

   public static void main(String[] args) {

      List<Person> people = Arrays.asList(
            new Person("Anurag"),
            new Person("Anurag"),
            new Person("Ravi")
      );

      Set<Person> distinct = new HashSet<Person>(people);

      Map<Integer, Long> result = new HashMap<Integer, Long>();

      for (Person p : distinct) {

         String mapped = p.name;                          // MAP
         Integer key = mapped.length();

         Long count = result.get(key);
         if (count == null) count = 0L;

         result.put(key, count + 1);                      // REDUCE + STORE
      }

      System.out.println(result);
```

```
    }
}
```

5 distinct() equals/hashCode + collect() (Map → Reduce → Store)

```
import java.util.*;
import java.util.stream.*;

public class DistinctEqualsMapReduceStore {

    static class Person {
        String name;

        Person(String name) { this.name = name; }

        @Override
        public boolean equals(Object o) {
            return o instanceof Person p && name.equals(p.name);
        }

        @Override
        public int hashCode() {
            return Objects.hash(name);
        }
    }

    public static void main(String[] args) {
        List<Person> people = List.of(
                new Person("Anurag"),
                new Person("Anurag"),
                new Person("Ravi")
        );

        Map<Integer, Long> result =
                people.stream()
                        .distinct()
                        .map(p -> p.name)                    // MAP
                        .collect(Collectors.groupingBy(
                                String::length,
                                Collectors.counting()        // REDUCE
                        ));

        System.out.println(result);
    }
}
```

6 distinct() immutable objects + collect() (Map → Reduce → Store)

```
import java.util.*;
import java.util.stream.*;

public class DistinctImmutableMapReduceStore {
   public static void main(String[] args) {
      List<String> values = List.of("Java", "Java", "Go", "Python");

      Map<Character, Long> result =
            values.stream()
                  .distinct()
                  .map(String::toUpperCase)                    // MAP
                  .collect(Collectors.groupingBy(
                        s -> s.charAt(0),
                        Collectors.counting()                  // REDUCE
                  ));

      System.out.println(result);
   }
}
```

---

7 map() transform objects + collect() (Map → Reduce → Store)

```
import java.util.*;

public class MapTransformMapReduceStore {
   public static void main(String[] args) {

      List<String> names = Arrays.asList("anurag", "amit", "ravi");
      Map<Integer, Long> result = new HashMap<Integer, Long>();

      for (String name : names) {

         String mapped = name.toUpperCase();                   // MAP
         Integer key = mapped.length();

         Long count = result.get(key);
         if (count == null) count = 0L;

         result.put(key, count + 1);                           // REDUCE + STORE
      }

      System.out.println(result);
   }
}
```

7 map() transform objects + collect() (Map → Reduce → Store)

```
import java.util.*;
import java.util.stream.*;

public class MapTransformMapReduceStore {
   public static void main(String[] args) {
      List<String> names = List.of("anurag", "amit", "ravi");

      Map<Integer, Long> result =
            names.stream()
                  .map(String::toUpperCase)                    // MAP
                  .collect(Collectors.groupingBy(
                        String::length,
                        Collectors.counting()                  // REDUCE
                  ));

      System.out.println(result);
   }
}
```

---

8 map() create new objects + collect() (Map → Reduce → Store)

```
import java.util.*;

public class MapNewObjectMapReduceStore {

   static class UserDTO {
      String name;
      UserDTO(String name) { this.name = name; }
   }

   public static void main(String[] args) {

      List<String> names = Arrays.asList("Anurag", "Amit", "Ravi");
      Map<Character, Long> result = new HashMap<Character, Long>();

      for (String name : names) {

         UserDTO dto = new UserDTO(name);                        // MAP
         Character key = dto.name.charAt(0);

         Long count = result.get(key);
         if (count == null) count = 0L;

         result.put(key, count + 1);                             // REDUCE + STORE
      }

      System.out.println(result);
```

```
    }
}
```

8 map() create new objects + collect() (Map → Reduce → Store)

```
import java.util.*;
import java.util.stream.*;

public class MapNewObjectMapReduceStore {

    static class UserDTO {
        String name;
        UserDTO(String name) { this.name = name; }
    }

    public static void main(String[] args) {
        List<String> names = List.of("Anurag", "Amit", "Ravi");

        Map<Character, Long> result =
            names.stream()
                .map(UserDTO::new)                          // MAP
                .collect(Collectors.groupingBy(
                    dto -> dto.name.charAt(0),
                    Collectors.counting()                   // REDUCE
                ));

        System.out.println(result);
    }
}
```

---

9 map() pure functions + collect() (Map → Reduce → Store)

```
import java.util.*;

public class MapPureFunctionMapReduceStore {

   static int square(int n) {
      return n * n;
   }

   public static void main(String[] args) {

      List<Integer> numbers = Arrays.asList(2, 3, 4);
      Map<Boolean, Integer> result = new HashMap<Boolean, Integer>();

      for (Integer n : numbers) {

         int mapped = square(n);                          // MAP
         Boolean key = (mapped > 10);

         Integer sum = result.get(key);
         if (sum == null) sum = 0;

         result.put(key, sum + mapped);                   // REDUCE + STORE
      }

      System.out.println(result);
   }
}
```

9 map() pure functions + collect() (Map → Reduce → Store)

```
import java.util.*;
import java.util.stream.*;

public class MapPureFunctionMapReduceStore {

   static int square(int n) {
      return n * n;
   }

   public static void main(String[] args) {
      List<Integer> numbers = List.of(2, 3, 4);

      Map<Boolean, Integer> result =
            numbers.stream()
                  .map(MapPureFunctionMapReduceStore::square)          // MAP
                  .collect(Collectors.groupingBy(
                        n -> n > 10,
                        Collectors.summingInt(Integer::intValue)       // REDUCE
                  ));

      System.out.println(result);
   }
}
```

---

10 flatMap() return Stream + collect() (Map → Reduce → Store)

```
import java.util.*;

public class FlatMapStreamMapReduceStore {
   public static void main(String[] args) {

      List<List<Integer>> lists = new ArrayList<List<Integer>>();
      lists.add(Arrays.asList(1, 2));
      lists.add(Arrays.asList(3, 4));

      Map<Boolean, Integer> result = new HashMap<Boolean, Integer>();

      for (List<Integer> inner : lists) {                    // FLATTEN
         for (Integer n : inner) {

            int mapped = n * 10;                             // MAP
            Boolean key = (mapped > 20);

            Integer sum = result.get(key);
            if (sum == null) sum = 0;

            result.put(key, sum + mapped);                   // REDUCE + STORE
         }
      }

      System.out.println(result);
   }
}
```

10 flatMap() return Stream + collect() (Map → Reduce → Store)

```
import java.util.*;
import java.util.stream.*;

public class FlatMapStreamMapReduceStore {
   public static void main(String[] args) {
      List<List<Integer>> lists = List.of(
            List.of(1, 2),
            List.of(3, 4)
      );

      Map<Boolean, Integer> result =
            lists.stream()
                  .flatMap(List::stream)
                  .map(n -> n * 10)                          // MAP
                  .collect(Collectors.groupingBy(
                        n -> n > 20,
                        Collectors.summingInt(Integer::intValue)  // REDUCE
                  ));
      System.out.println(result);
   }
}
```

11 flatMap() split data + collect() (Map → Reduce → Store)

```
import java.util.*;

public class FlatMapSplitMapReduceStore {
   public static void main(String[] args) {

      List<String> sentences = Arrays.asList("Java Stream", "Flat Map");
      Map<Integer, Long> result = new HashMap<Integer, Long>();

      for (String sentence : sentences) {

         String[] words = sentence.split(" ");                    // SPLIT

         for (int i = 0; i < words.length; i++) {

            String mapped = words[i].toLowerCase();               // MAP
            Integer key = mapped.length();

            Long count = result.get(key);
            if (count == null) count = 0L;

            result.put(key, count + 1);                           // REDUCE + STORE
         }
      }

      System.out.println(result);
   }
}
```

11 flatMap() split data + collect() (Map → Reduce → Store)

```
import java.util.*;
import java.util.stream.*;

public class FlatMapSplitMapReduceStore {
   public static void main(String[] args) {
      List<String> sentences = List.of("Java Stream", "Flat Map");

      Map<Integer, Long> result =
            sentences.stream()
                  .flatMap(s -> Arrays.stream(s.split(" ")))
                  .map(String::toLowerCase)                       // MAP
                  .collect(Collectors.groupingBy(
                        String::length,
                        Collectors.counting()                     // REDUCE
                  ));
      System.out.println(result);
   }
```

```
}
```

---

12 flatMap() expand collections + collect() (Map → Reduce → Store)

```
import java.util.*;

public class FlatMapExpandMapReduceStore {
   public static void main(String[] args) {

      Map<String, List<String>> data = new HashMap<String, List<String>>();
      data.put("A", Arrays.asList("x", "y"));
      data.put("B", Arrays.asList("z"));

      Map<Integer, Long> result = new HashMap<Integer, Long>();

      for (List<String> list : data.values()) {                    // FLATTEN
         for (String value : list) {

            String mapped = value.toUpperCase();                    // MAP
            Integer key = mapped.length();

            Long count = result.get(key);
            if (count == null) count = 0L;

            result.put(key, count + 1);                             // REDUCE + STORE
         }
      }

      System.out.println(result);
   }
}
```

12 flatMap() expand collections + collect() (Map → Reduce → Store)

```
import java.util.*;
import java.util.stream.*;

public class FlatMapExpandMapReduceStore {
   public static void main(String[] args) {
      Map<String, List<String>> data = Map.of(
            "A", List.of("x", "y"),
            "B", List.of("z")
      );

      Map<Integer, Long> result =
            data.values().stream()
                  .flatMap(Collection::stream)
                  .map(String::toUpperCase)                    // MAP
                  .collect(Collectors.groupingBy(
                        String::length,
                        Collectors.counting()                  // REDUCE
                  ));

      System.out.println(result);
   }
}
```

---

13 sorted() natural order + collect() (Map → Reduce → Store)

```
import java.util.*;

public class SortedNaturalMapReduceStore {
   public static void main(String[] args) {

      List<Integer> numbers = new ArrayList<Integer>(
            Arrays.asList(5, 1, 3));

      Collections.sort(numbers);                                  // SORT

      Map<Boolean, Integer> result = new HashMap<Boolean, Integer>();

      for (Integer n : numbers) {

         int mapped = n * 2;                                      // MAP
         Boolean key = (mapped > 5);

         Integer sum = result.get(key);
         if (sum == null) sum = 0;

         result.put(key, sum + mapped);                           // REDUCE + STORE
      }

      System.out.println(result);
   }
}
```

13 sorted() natural order + collect() (Map → Reduce → Store)

```
import java.util.*;
import java.util.stream.*;

public class SortedNaturalMapReduceStore {
   public static void main(String[] args) {
      List<Integer> numbers = List.of(5, 1, 3);

      Map<Boolean, Integer> result =
            numbers.stream()
                  .sorted()
                  .map(n -> n * 2)                                // MAP
                  .collect(Collectors.groupingBy(
                        n -> n > 5,
                        Collectors.summingInt(Integer::intValue)  // REDUCE
                  ));

      System.out.println(result);
```

```
    }
}
```

---

14 sorted() custom comparator + collect() (Map → Reduce → Store)

```
import java.util.*;

public class SortedCustomMapReduceStore {
    public static void main(String[] args) {

        List<String> names = new ArrayList<String>(
                Arrays.asList("Anurag", "Ravi", "Amit"));

        Collections.sort(names, Collections.reverseOrder());    // CUSTOM SORT

        Map<Character, Long> result = new HashMap<Character, Long>();

        for (String name : names) {

            String mapped = name.toUpperCase();                 // MAP
            Character key = mapped.charAt(0);

            Long count = result.get(key);
            if (count == null) count = 0L;

            result.put(key, count + 1);                         // REDUCE + STORE
        }

        System.out.println(result);
    }
}
```

14 sorted() custom comparator + collect() (Map → Reduce → Store)

```
import java.util.*;
import java.util.stream.*;

public class SortedCustomMapReduceStore {
   public static void main(String[] args) {
      List<String> names = List.of("Anurag", "Ravi", "Amit");

      Map<Character, Long> result =
            names.stream()
                  .sorted(Comparator.reverseOrder())
                  .map(String::toUpperCase)                    // MAP
                  .collect(Collectors.groupingBy(
                        s -> s.charAt(0),
                        Collectors.counting()                  // REDUCE
                  ));

      System.out.println(result);
   }
}
```

---

15 sorted() multi-field + collect() (Map → Reduce → Store)

```
import java.util.*;

public class SortedMultiFieldMapReduceStore {

   static class Employee {
      String name;
      int age;

      Employee(String name, int age) {
         this.name = name;
         this.age = age;
      }
   }

   public static void main(String[] args) {

      List<Employee> employees = new ArrayList<Employee>();
      employees.add(new Employee("Anurag", 30));
      employees.add(new Employee("Anurag", 25));
      employees.add(new Employee("Ravi", 35));

      Collections.sort(employees, new Comparator<Employee>() {
         public int compare(Employee e1, Employee e2) {

            int nameCompare = e1.name.compareTo(e2.name);
            if (nameCompare != 0) return nameCompare;

            return e1.age - e2.age;
         }
      });

      Map<String, Integer> result = new HashMap<String, Integer>();

      for (Employee e : employees) {

         int mapped = e.age;                                    // MAP
         String key = (mapped >= 30) ? "SENIOR" : "JUNIOR";

         Integer sum = result.get(key);
         if (sum == null) sum = 0;

         result.put(key, sum + mapped);                         // REDUCE + STORE
      }

      System.out.println(result);
   }
```

```
}
```

15 sorted() multi-field + collect() (Map → Reduce → Store)

```
import java.util.*;
import java.util.stream.*;

public class SortedMultiFieldMapReduceStore {

   static class Employee {
      String name;
      int age;

      Employee(String name, int age) {
         this.name = name;
         this.age = age;
      }
   }

   public static void main(String[] args) {
      List<Employee> employees = List.of(
            new Employee("Anurag", 30),
            new Employee("Anurag", 25),
            new Employee("Ravi", 35)
      );

      Map<String, Integer> result =
            employees.stream()
                  .sorted(Comparator
                        .comparing((Employee e) -> e.name)
                        .thenComparingInt(e -> e.age))
                  .map(e -> e.age)                                  // MAP
                  .collect(Collectors.groupingBy(
                        age -> age >= 30 ? "SENIOR" : "JUNIOR",
                        Collectors.summingInt(Integer::intValue)  // REDUCE
                  ));

      System.out.println(result);
   }
}
```

16 peek() logging + collect() (Map → Reduce → Store)

```
import java.util.*;

public class PeekLoggingMapReduceStore {
   public static void main(String[] args) {

      List<Integer> list = Arrays.asList(1, 2, 3);
      Map<Boolean, Integer> result = new HashMap<Boolean, Integer>();

      for (Integer n : list) {

         System.out.println("Log: " + n);                          // PEEK

         int mapped = n * 10;                                      // MAP
         Boolean key = (mapped > 20);

         Integer sum = result.get(key);
         if (sum == null) sum = 0;

         result.put(key, sum + mapped);                            // REDUCE + STORE
      }

      System.out.println(result);
   }
}
```

16 peek() logging + collect() (Map → Reduce → Store)

```
import java.util.*;
import java.util.stream.*;

public class PeekLoggingMapReduceStore {
   public static void main(String[] args) {
      Map<Boolean, Integer> result =
            List.of(1, 2, 3).stream()
                  .peek(n -> System.out.println("Log: " + n))
                  .map(n -> n * 10)                                 // MAP
                  .collect(Collectors.groupingBy(
                        n -> n > 20,
                        Collectors.summingInt(Integer::intValue)  // REDUCE
                  ));

      System.out.println(result);
   }
}
```

17 peek() debugging + collect() (Map → Reduce → Store)

```
import java.util.*;

public class PeekDebugMapReduceStore {
   public static void main(String[] args) {

      List<Integer> list = Arrays.asList(10, 20, 30);
      Map<Boolean, Long> result = new HashMap<Boolean, Long>();

      for (Integer n : list) {

         if (n == 20) {
            System.out.println("Debug hit: " + n);                    // PEEK
         }

         int mapped = n + 1;                                          // MAP
         Boolean key = (mapped > 20);

         Long count = result.get(key);
         if (count == null) count = 0L;

         result.put(key, count + 1);                                  // REDUCE + STORE
      }

      System.out.println(result);
   }
}
```

17 peek() debugging + collect() (Map → Reduce → Store)

```
import java.util.*;
import java.util.stream.*;

public class PeekDebugMapReduceStore {
   public static void main(String[] args) {
      Map<Boolean, Long> result =
            List.of(10, 20, 30).stream()
                  .peek(n -> {
                     if (n == 20) System.out.println("Debug hit: " + n);
                  })
                  .map(n -> n + 1)                                    // MAP
                  .collect(Collectors.groupingBy(
                        n -> n > 20,
                        Collectors.counting()                         // REDUCE
                  ));

      System.out.println(result);
   }
```

```
}
```

---

18 peek() metrics + collect() (Map → Reduce → Store)

```
import java.util.*;

public class PeekMetricsMapReduceStore {
   public static void main(String[] args) {

      List<Integer> list = Arrays.asList(5, 10, 15);
      int counter = 0;

      Map<Boolean, Integer> result = new HashMap<Boolean, Integer>();

      for (Integer n : list) {

         counter++;                                          // METRICS (peek equivalent)

         int mapped = n * 2;                                 // MAP
         Boolean key = (mapped > 20);

         Integer sum = result.get(key);
         if (sum == null) sum = 0;

         result.put(key, sum + mapped);                      // REDUCE + STORE
      }
      System.out.println("Processed: " + counter);
      System.out.println(result);
   }
}
```

18 peek() metrics + collect() (Map → Reduce → Store)

```
import java.util.*;
import java.util.concurrent.atomic.AtomicInteger;
import java.util.stream.*;

public class PeekMetricsMapReduceStore {
   public static void main(String[] args) {
      AtomicInteger counter = new AtomicInteger();

      Map<Boolean, Integer> result =
            List.of(5, 10, 15).stream()
                  .peek(n -> counter.incrementAndGet())
                  .map(n -> n * 2)                                    // MAP
                  .collect(Collectors.groupingBy(
                        n -> n > 20,
                        Collectors.summingInt(Integer::intValue) // REDUCE
                  ));
      System.out.println("Processed: " + counter.get());
      System.out.println(result);
   }
}
```